45. Host Network Performance Tuning Intro

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Phase: Phase 11 — Observability and troubleshooting
Estimated time: 1–2 days
Slug: `45-performance-tuning`

Learning objectives

- Inspect RX/TX errors and drops
- Know ephemerial port exhaustion exists
- Tune carefully with evidence
- Relate to ulimit/somaxconn

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

```
ip -s link
netstat -s | head -50                  # or ss -s
cat /proc/sys/net/ipv4/ip_local_port_range   # Linux
sysctl net.core.somaxconn
ss -s
```

High `CLOSE-WAIT`/`TIME-WAIT`, port range exhaustion, NIC errors — measure first, then tune.

Common mistakes

- sysctl cargo cult from blogs
- Raising limits hiding leaks
- Ignoring NIC hardware offload/errors

Exercises

Work in `exercises/45-performance-tuning/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/45-performance-tuning/` until after an honest attempt.

Self-check

1. Where see interface drops?
2. Ephemeral ports why?
3. Measure before tune?

Further reading

- https://www.kernel.org/doc/Documentation/networking/ip-sysctl.txt